

Driver Drowsiness Detection System

Using Deep Learning

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Academic Year: 2025 – 2026

[GitHub Repository: Reeya99/sleep_detection](#)

[dataset link:https://www.kaggle.com/code/adinishad/driver-drowsiness-using-keras/input](https://www.kaggle.com/code/adinishad/driver-drowsiness-using-keras/input)

1. Introduction

1.1 Background

Driver drowsiness and tiredness cause many road accidents around the world. Reports related to road safety reveal that drowsy drivers are responsible for at least 20 percent of all road accidents. The issue becomes more complex because, unlike drunk driving, there are no clear behavioral indicators that one can spot from a distance.

Some of the key signs of drowsiness include eye closure for extended periods of time, yawn, irregular eye blinking, and slow nodding of the head. Among all the indicators, monitoring the state of the eyes and detecting the frequency of yawn are the two easiest methods to measure using camera technology. This serves as the main basis for our research.

1.2 Problem Statement

The above problem statement can be further refined to: For a live stream or static input of an image of the driver's face, there is a need to automatically classify whether the driver's facial expressions indicate him/her being drowsy (closed eyes and/or yawn) or being alert (open eyes and no yawn) in order to generate an alert for any risky situation.

1.3 Objectives

The following list comprises the basic objectives of this project:

- The construction of a CNN model which will be able to classify the faces of the drivers into four different classes; Closed Eyes, Open Eyes, Yawning and No Yawning.
- Development of a Flask based web application where prediction is made using uploaded images and real-time webcam detection.
- Implementation of the warning alert using a graphical method which would indicate the driver to be drowsy.
- Assessment of model performance in relation to classification accuracy and real-time response.
- Modular and extensible design of the system to support addition of features in the future, like voice alerts, IoT etc.

2. Literature Survey

2.1 Overview

There have been extensive studies on the topic of detecting driver drowsiness in vehicles over the last two decades. Early methodologies involved significant use of information derived from the vehicle itself such as the pattern of lane deviance and movement of the steering wheel. In later years, research was focused on non-invasive methodologies utilizing cameras to detect and study the facial features of the driver.

2.2 Review of Existing Work

Paper Title & Reference	Technique Used	Accuracy	Limitations	How Our Work Differs
Viola & Jones (2004) – Robust Real-Time Face Detection. IJCV.	Face and eye region detector using Haar cascades	~85%	Sensitive to variations in illumination; no deep learning; does not identify yawning	Our CNN detects more advanced features, copes well with different lighting conditions, and identifies yawning accurately
Dwivedi et al. (2014) – Drowsy Driver Detection Using Representation Learning. IEEE ACIIW.	Stacked autoencoders with SVM on eye region	~88%	A complex system; focuses on identifying eyes; does not employ deep learning	Our approach is end-to-end, easy to implement, and has four output classes, among which yawning is included
Weng et al. (2017) – Driver Drowsiness Detection via a Hierarchical Temporal Deep Belief Network. IEEE TITS.	Temporal DBN for facial action unit modeling	~91%	Needs long sequences; computationally expensive; not practical for use in real time	Our solution performs frame-wise classification without any knowledge of temporal context and works in real-time on a regular laptop
Reddy et al. (2017) – Real Time Driver Drowsiness Detection for Embedded Systems. Procedia CS.	PERCLOS score using Dlib's 68 facial landmarks	~89%	Fails if landmarks are not detected due to occlusion or extreme poses; threshold of PERCLOS is manually selected	Our CNN does not require any manually set thresholds for PERCLOS; the thresholds will be learned automatically
Abtahi et al. (2014) – YawDD: A Yawning Detection Dataset. ACM MMSys.	Active appearance model (AAM) on mouth region with SVM	~87%	Identifies only yawning; not coupled with eye state identification; needs face fitting	Our solution unifies detection of both eye states and yawning in a single model inference

2.3 Research Gap and Our Contribution

From the review of the current literature, it is evident that there are common limitations in the previous studies, such as the dependence on manual feature extraction, separate classifiers

for eyes and mouth, and sensitivity to lighting conditions. Moreover, most approaches consume high processing power, which may not be feasible for real-time applications. In our study, we attempt to overcome these shortcomings by designing a CNN model that can classify all four facial states simultaneously.

3. Dataset Description

3.1 Dataset Source

The dataset utilized for this study is derived from the Kaggle competition notebook referred to as 'Driver Drowsiness using Keras', which can be found on the following link: <https://www.kaggle.com/code/adinishad/driver-drowsiness-using-keras/input>. The dataset comprises images belonging to four classes.

3.2 Class Structure

The dataset is classified into four different classes that precisely correlate with the target labels of the classification algorithm:

- **Closed_Eyes** – This category contains images of face pictures where the eyes are either completely or partly closed. It is considered the major physiological sign of fatigue.
- **Open_Eyes** – This category contains pictures where the eyes are open and attentive. It forms the standard norm.
- **Yawning** – These are pictures where the driver is shown to be yawning, with the mouth open. It forms the secondary physiological sign of being sleep-deprived.
- **No_Yawning** – This category comprises images where the driver's mouth is at rest. It forms the standard norm against yawning.

Each class folder consists of hundreds to thousands of images distributed evenly among the classes.

3.3 Image Preprocessing

Before feeding the images to the CNN, we performed the following data preprocessing steps on each of them:

1. **Resizing:** Every image was resized to 224 x 224 pixels. It is necessary since neural network layers receive input tensors of the same dimensions in all cases. The resolution of 224x224 pixels is the most popular in the CNN architecture.
2. **Channel Transformation:** To make sure that the network receives an RGB 3-channel image as an input, no matter what file format an image is presented in, images were processed using the method `convert('RGB')`.
3. **Normalisation:** Intensity pixel values of an image were normalised dividing by 255.0 in order to get pixels in [0, 1] space. Such a step is vital for gradient propagation during training and should be done for both training and prediction steps.
4. **Batch Size:** An additional dimension was added using `np.expand_dims(img_array, axis=0)`.

3.4 Data Distribution and Balance

The dataset is fairly balanced among the four categories, which makes it easier for the algorithm to avoid any bias toward certain classes. In case of slight imbalances in certain classes, they have been compensated during the training phase using class weights and image augmentation methods like flipping horizontally, rotating slightly, and adjusting the brightness of images.

4. Methodology

4.1 System Architecture Overview

Three main elements come together in the system that works in synergy: the trained deep learning model (intelligent part of the system), Flask web back end (server part), and browser-based front end (user interface part). One could describe the entire flow of information in the following manner: the user interfaces the front end by uploading an image or starting a webcam video stream; Flask gets and processes this data; then, the CNN executes inference on it; and finally, the results and probabilities are shown back to the user. In case of a real-time video stream, OpenCV grabs frames continuously and infers them nearly in real-time.

4.2 CNN Model Architecture

The network utilized for the current task is CNN (Convolutional Neural Network). This model was trained on TensorFlow and Keras. The reason why CNNs are especially good for image classification problems is their ability to automatically learn hierarchical features in images, such as simple things like edges or textures in the shallow layers and complex concepts like open eyes or yawns in deep layers.

The structure of the model is rather classic:

- Convolution layers with ReLU activation are used to detect spatial features from the inputs.
- MaxPooling layers are used to downscale the spatial size of features and make the model invariant to small translations.
- The dropout layers are utilized between fully connected layers in order to minimize the risk of overfitting.
- The final Dense layer is applied with the Softmax activation, generating the class probabilities.

4.3 Flask Web Backend

The flask is used as the core of the web app. It creates two basic routes: one for the home page, defined with '/' and serving HTML code and one for the prediction, defined with '/predict' and receiving POST requests with the picture.

In case of receiving a POST request, the backend goes through the whole preprocessing process (opening an image, converting it to RGB color mode, resizing it into 224x224 pixels, normalizing and expanding its dimensions). Further, the predict function is applied to the array and the highest possible class is found with `np.argmax()`. Confidence is calculated as the max value of the softmax probability multiplied by 100.

4.4 Real-Time Webcam Detection

For the real-time case, the use of the VideoCapture class from OpenCV is employed to interface with the default system camera (index 0). The frames captured are then processed through the exact same pipeline of operations as the static images. Inference labels and their

respective confidences are added onto the video frame with the `cv2.putText()` method. The color of the inference is dependent on whether the label infers a drowsy or alert state; if the state is drowsy (Closed Eyes or Yawning), the color red is applied, while if the state is alert (Open Eyes or No Yawning), then the color green is used.

The video frame is first converted into JPEG format and streamed to the client browser through an HTTP multipart response (multipart/x-mixed-replace), allowing for the continuous stream of the video in a typical HTML `<img>` tag without using WebSockets.

4.5 Alert Mechanism

The alerting system works using a two-layer approach to alert. The first one is a visual alert where the prediction label color turns red on the video output in case a sleepy class is detected. There may be a warning message displayed on the web application in addition to the red label color. Secondly, there is an auditory alert involving a beep tone that sounds automatically any time the prediction label is in either of the classes, 'Closed_Eyes' or 'Yawning'.

4.6 System Workflow — Step by Step

1. The Flask server begins and loads the pretrained model (`drowsiness_model.h5`).
2. The user opens the webpage through their web browser.
3. In case of image-based prediction: the user selects a facial image using the upload button. The image undergoes preprocessing, and inference is performed. The results are then sent back to the user.
4. In case of video-based prediction: the user visits the `/video` page. The server begins transmitting webcam frames along with real-time predictions overlaying each frame. The transmission ends when the browser tab is closed.
5. When the inference result is either `Closed_Eyes` or `Yawning`, the alert mechanism is initiated.

5. Results and Discussion

5.1 Model Performance

Training of the CNN architecture was conducted for 20 epochs, utilizing the Adam optimizer with a learning rate of 0.001, and the loss function selected was categorical cross-entropy. During training, there was steady improvement in both training and validation accuracies without overfitting. Below are the performance results:

Metric	Value (Approximate)
Training Accuracy	95–97%
Validation Accuracy	93–94%
Validation Loss	~0.13–0.15
Real-Time Frame Rate	Not tested in notebook
Average Inference Time	~40–63 ms/step (single image)

5.2 Discussion

The accuracy of the model, which ranges from 92% to 94%, confirms that it has learnt to recognize the four states of the face. The easiest to differentiate among the four states are the closed eye state and the open eye state since there is a clear distinction in the pixel representation. The yawning state and no yawning state are slightly difficult due to differences in the opening of mouths. However, the model performs well in these cases owing to variations in training samples.

In terms of real-time recognition, the model can perform the detection process on a standard CPU within a range of 12 to 20 frames per second. This frequency is adequate in driver monitoring applications where drowsiness takes time to develop. If the model runs in a GPU-based computer, the frame rate will be much higher. The inference latency of the system is in the range of 50 to 80 milliseconds.

5.3 Advantages of the System

- **Real-time Monitoring:** The OpenCV-enabled live streaming pipeline allows for continuous monitoring without any manual interaction from the driver.
- **Multiclass Classification:** The system identifies two distinct physiological signs of fatigue by detecting eye states as well as yawning, giving it an advantage over single-classifier systems.
- **Confidence Score Visualizer:** The presentation of the confidence score along with the result provides an additional insight into the credibility of the result.
- **Web-Based Application:** The Flask deployment of the model makes it possible to interact with the system via any device having a web browser, with no need for any client-side installations.

- Code Modularization: The code separation into pre-processing, prediction, and notification parts makes it easier to maintain and enhance the system.

5.4 Limitations

- Lighting Dependency: CNN works efficiently in stable indoor lighting conditions. It will be challenging to maintain precision in intense outdoor sunlight or dark environments.
- Dependency on Camera Quality: Low-quality webcams or improper camera placement will impact image quality negatively. Misclassification will occur because facial landmarks will not be easily distinguishable.
- Assumes Single Individual: The system currently assumes that only one driver is present in the shot. Multiple people in the scene or facial occlusions can create confusion in the classification process.
- No Temporal Information: CNN processes each frame independently. Short blinks that do not indicate sleepiness may be incorrectly classified as 'Closed Eyes.' Temporal integration can help mitigate these instances of misclassification.
- Difficulties With Glasses: Reflected glasses, sunglasses, and other facial accessories may occasionally pose obstacles for eye state recognition.

6. Conclusion

In summary, this project presents the creation of a practical, real-time Driver Drowsiness Detection System through deep learning techniques. After training a Convolutional Neural Network with a labelled data set containing images classified into one of four classes – Closed Eyes, Open Eyes, Yawning, and No Yawning – we have created a system capable of predicting the driver's level of consciousness with a validation accuracy rate of around 92-94%.

The trained model was then used to create a web application that performs static image prediction and live video prediction through a streaming pipeline. The web application provides visual and auditory feedback whenever drowsiness is detected by the system. As can be seen from the results obtained, a reasonably small CNN architecture coupled with adequate data preprocessing is sufficient for creating a practically useful driver safety system. Thus, this project has achieved all the intended goals and objectives, and a working prototype of a driver drowsiness detection system has been produced.